Forecasts vs. Reality: Lieberman-Warner Permit Prices and Real Cap-and-Trade Markets

Xiwen Zhang*

University of Illinois Urbana-Champaign, Gies College of Business

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Abstract

In 2008, forecasts of the Lieberman-Warner Climate Security Act predicted carbon permit prices of roughly $17 to $81 per metric ton by 2015. This paper compares those estimates analyzed by Buckley and Mityakov (2009) with actual permit prices from the EU Emissions Trading System, California, and the Regional Greenhouse Gas Initiative. Every program's allowance price came in below the minimum forecast price both in 2015 and 2020, with only the EU reaching the 2030 forecast range. At the observed prices, the expected effects on household consumption and residential electricity prices were too small to detect in state-level data. Comparing neighboring states with treated states showed that electricity price premiums followed natural gas prices more than permit costs, suggesting that fuel markets the 2008 forecasts did not anticipate affected household energy costs more than carbon permits did.

1. Introduction

In 2007, the U.S. Senate considered the Lieberman-Warner Climate Security Act (S.2191), which is a cap-and-trade bill that would have covered approximately 86% of emissions as Buckley (2009) estimates. The bill would have aimed to lower the cap each year until 2050. However, this bill never passed. A vote to move it forward in June 2008 gained 48 votes, short of the 60 needed. Before this vote, several government, academic, and industry groups had

* University of Illinois Urbana-Champaign, Gies College of Business. Email: xiwenz4@illinois.edu. This research received no external funding. I thank Dr. Bryan Buckley for his guidance and for suggesting the neighboring state comparison.

provided forecasts of what this bill would cost. Buckley and Mityakov (2009), which also appears as Chapter 3 of Buckley's (2009) dissertation, compared seven of these forecasts and estimated that the bill would result in a decrease in household consumption of approximately 1%. Permit prices are an important factor in forecasts like this one, as higher permit prices pass the cost of energy and goods down to consumers. Since the Lieberman-Warner Act never passed into law, its forecasts could never be checked against real outcomes. However, the European Union (EU), California, and the Regional Greenhouse Gas Initiative (RGGI) have all implemented cap-and-trade policies since then. This preliminary analysis asks how well the 2008 forecasts predicted allowance prices, what explains the gap between the forecasts and reality, whether the programs' costs actually affected households in measurable ways, and what drove electricity price differences between programs and neighboring states.

2. Methodology

The forecasted prices are derived from Table 5 of Buckley and Mityakov's (2009) paper, which reported carbon allowance prices using 2007 dollars per metric ton of carbon dioxide for each study and scenario. The table was entered by hand into an Excel spreadsheet, exported as a .csv file, and cleaned using the pandas library to reshape the DataFrame so that each row has one forecast per year. There were two studies that reported their short-term forecasts in different years than the table's column headers, so the ACCF/NAM and EIA values were recorded in 2014 and 2020, respectively.

Actual auction prices were obtained from each program's own auction results, publicly available on the internet. EU prices are primary auction clearing prices from the European Energy Exchange (EEX), with data starting from 2012. California auction prices are quarterly settlement prices obtained from the California Air Resources Board (CARB) PDF files, with data

obtained starting from November 2012. RGGI prices are quarterly auction clearing prices, with data beginning from 2008.

In order to compare these prices with the forecasts, all actual prices were converted to 2007 dollars per metric ton in three steps. First, EU prices were converted from euros to dollars at the daily exchange rate on each auction date. The data for the financial exchange rate was obtained from the Federal Reserve Economic Data (FRED) website. Next, RGGI data had to be converted from short tons to metric tons. Lastly, all prices were adjusted for inflation to 2007 dollars using the historical Consumer Price Index obtained from the Federal Reserve Bank of St. Louis on the FRED website. Prices were converted before being averaged, since exchange rates vary throughout the year, and then averaged by program and year. All work was done in Python using pandas, and every source, date range, and data manipulation decision is recorded in the Jupyter Notebook file and/or the project's data log.

Household data was obtained from the Bureau of Economic Analysis (BEA) which reports Personal Consumption Expenditure (PCE) per capita and population by state. Residential electricity prices, generation, and power-sector CO2 emissions data was obtained from the Energy Information Administration's website. Consumption data covers 2000 through 2025, and electricity data covers 2001 to 2024. All dollar values were normalized to 2007 dollars using CPI information. Three groups were created: the nine core RGGI states, California, and never-treated states as a baseline. New Jersey, Virginia, and Washington were excluded from all groups because each was covered by a program for only part of the period. Two checks were made, one of which was the expected effect of the permits. This was done by dividing the permit cost by the consumption or by the electricity price. The other check was finding the smallest detectable

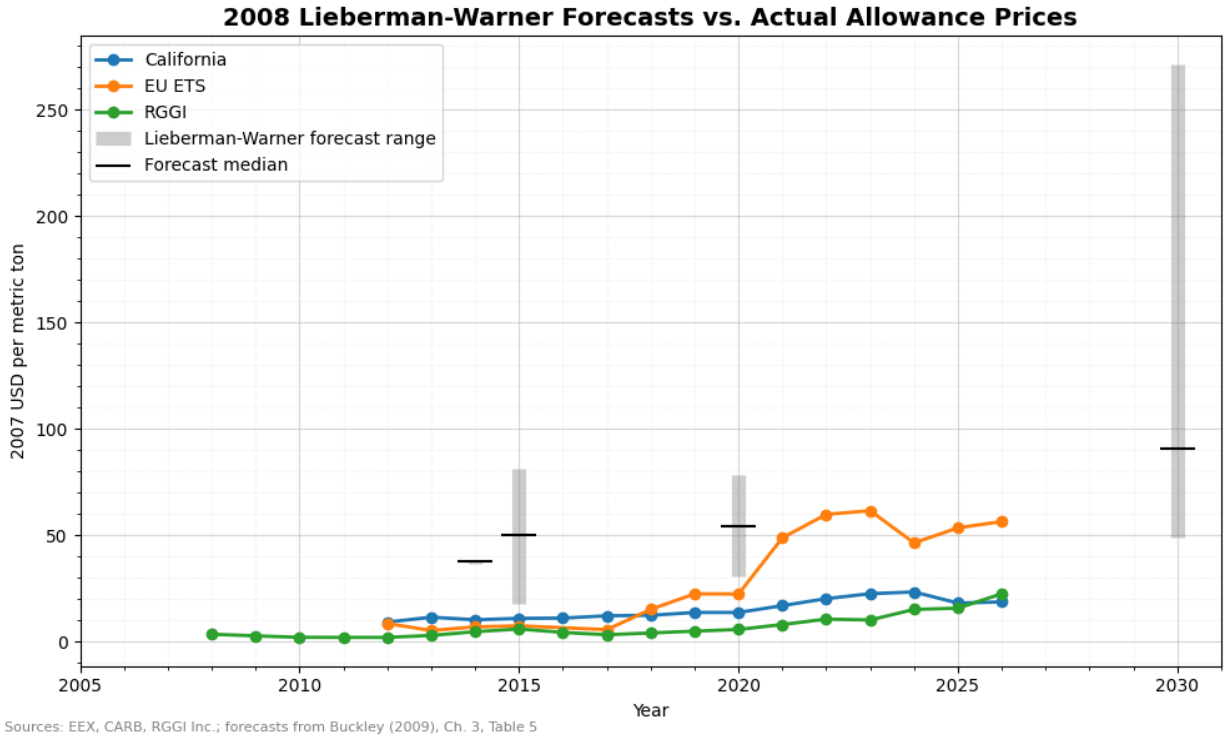
effect, which was estimated from pre-period noise in the gap between treated and never-treated states.

For Section 6, each program was also compared with its bordering never-treated states, both visually and with a difference-in-differences regression using state and year fixed effects.

3. Results and findings

The 2015 Lieberman-Warner forecasts ranged from $17.43 to $81.13 per metric ton with a median of $50.37. Actual prices in California, the EU ETS, and RGGI differed substantially from these forecasts. The actual price for California in 2015 was $10.88, for the EU it was $7.38, and for RGGI it was $5.88. Every program came in below the minimum price predicted by the forecast. This trend followed for the 2020 Lieberman-Warner forecasts as well, with all actual prices coming in at $13.65 (California), $22.32 (EU), and $5.66 (RGGI), well below the minimum price forecast of $30.84 per metric ton (this is a single forecast from EIA). However, after 2020, the EU rose sharply until 2023 to approximately $61.51 per metric ton while RGGI and California stayed at $10.12 and $22.47 per metric ton in 2023, respectively.

Figure 1. Actual vs. Prediction



Source: Forecasts from Buckley and Mityakov (2009), Table 5. Actual prices from EEX (EU ETS), CARB (California), and RGGI, Inc. Converted to 2007 dollars per metric ton using FRED series CPIAUCSL and DEXUSEU. Author's calculations.

Looking forward to 2030, the forecast range is from $49.03 to $271.27 per metric ton, with a median of $90.83 per metric ton. As of 2026, the EU has already surpassed the minimum forecasted price, although California and RGGI are still well below the minimum forecasted price, with a combined average price of $20.54 per metric ton while the EU is hovering around $56.41 per metric ton (more than double the average price of the US policies).

4. Why the gap?

There are several reasons for the difference in forecast numbers and actual numbers. As Murray and Maniloff (2015) state, the widening gap between emissions and generation may be attributed to the "decarbonization of power generation." In other words, as more power plants switch over to fuels that are cheaper and produce less carbon dioxide, the demand for these carbon permits drops, and thus the price falls. Romm (2008) adds that the forecasts did not take

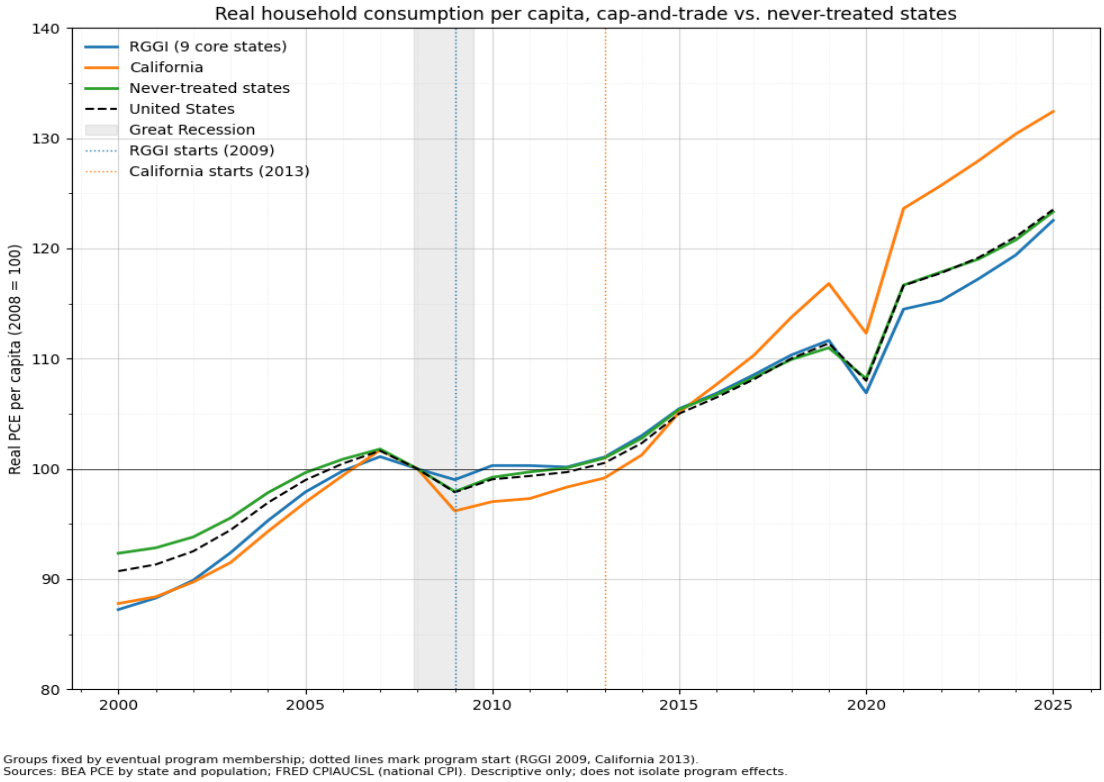
into consideration new technologies that could produce electricity at a lower cost and with fewer emissions. However, the EU offers a different side of the story. It connects directly to the cost drivers Buckley and Mityakov (2009) identified.

The EU ETS has faced a large surplus of permits. The 2008 economic crisis decreased emissions below what the initial cap had planned for, and a high inflow of international allowances added to the supply. The international credits could only be used until 2020 (European Commission 2026). Companies banked the extra permits, and by 2013, surplus permits had exceeded 2.1 billion (European Commission 2026). To combat this, the EU created the Market Stability Reserve, which saw 2.5 billion allowances cancelled in 2023 (European Commission 2026). The Commission also tightened the cap on emissions through the "Fit for 55" legislation, which targets a reduction in emissions of at least 55% by 2030, compared to 1990 levels (European Commission 2021). Since international credits could only be used until 2020 and the "Fit for 55" legislation was going into effect in 2021, indicating a tighter cap of emissions, the expectation of future scarcity caused companies to bank more of the allowances and prices increased suddenly. The EU's experience suggests that offsets, banking, and tight policies shaped permit prices as Buckley and Mityakov (2009) expected. After the oversupply ended, the EU ETS permit prices moved into the low end of the 2008 projections for 2030.

5. Household effects and detectability

As Figure 2 shows, treated states do not diverge from never-treated states when their programs start. Before 2008, California and RGGI states grew faster than never-treated states, so the pre-trends were not parallel either. All the lines intersect in 2008 because each one is indexed to that year. Additionally, California's higher consumption and consumption growth may also reflect the technology boom around the same period as the cap-and-trade policies were in effect.

Figure 2. Consumption



Source: BEA, personal consumption expenditures per capita and population by state. Deflated with FRED series CPIAUCSL. Groups are fixed by eventual program membership; New Jersey, Virginia, and Washington are excluded. Author's calculations.

Electricity prices are where the effects of rising allowance prices should appear first. Both programs cover power plants, and RGGI covers only power plants, so permit costs impact electricity generation directly. Utility costs will eventually reach consumers through retail rates which regulators adjust, often with a delay.

However, these effects could not be reliably detected upon analysis. This could be due to many factors, such as low permit prices, narrow coverage of the programs, or large amounts of noise from outside economic effects. Low permit prices mean the burden per household is small relative to normal consumption, about $7 to $9 per person per year in RGGI states and $110 to $115 in California, in 2007 dollars. This fact in addition to the narrow coverage of the programs and the large amounts of noise from other economic factors like state policies, housing booms, tech growth, or the COVID-19 pandemic made detecting a minute change improbable.

Additionally, the selected design itself was weak. Because electricity data starts in 2001, RGGI only has 8 years of baseline before treatment. Furthermore, never-treated states were already moving differently from treated states even before the programs started.

Table 1. Detectability

Outcome	Program	Expected Effect	Smallest Detectable	Was it detectable?
Consumption	RGGI	0.02%	0.19%	No
Consumption	California	0.3%	1.3%	No
Electricity Price	RGGI	1.0%	4.2%	No
Electricity Price	California	2.0%	4.2%	No

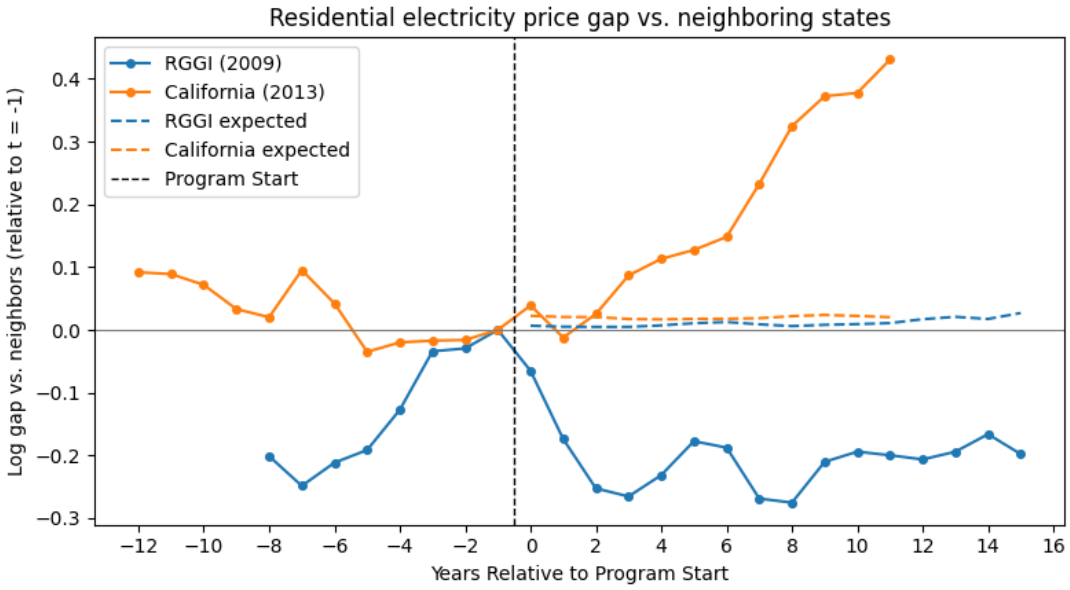
Source: Author's calculations using BEA (consumption and population), EIA (residential electricity prices, generation, and power-sector CO2), CARB (17 CCR § 95841 allowance budgets), RGGI, Inc. (adjusted caps and auction prices), and FRED (CPIAUCSL).

At the observed permit prices, the costs per household were too small relative to normal state-to-state variations to be measured with state-level data. The roughly 1% consumption loss Buckley and Mityakov (2009) estimated assumed higher permit prices and a much broader 86% coverage of emissions under the Lieberman-Warner Act.

6. What moved electricity prices?

To reduce national-level noise, each program was compared to its bordering never-treated states. RGGI was compared to Pennsylvania and West Virginia. Oregon, Nevada, and Arizona were used as comparison states for California.

Figure 3. Residential electricity price premium vs. neighboring states



Source: EIA, average residential retail electricity price by state, 2001–2024. Expected paths use auction prices from CARB and RGGI, Inc. and EIA power-sector CO2 emissions and net generation by state. Deflated to 2007 dollars with FRED series CPIAUCSL. Neighbor groups: Pennsylvania and West Virginia (RGGI); Oregon, Nevada, and Arizona (California). Each line is normalized to the year before the program started. Author's calculations.

Figure 3 shows the log gap in residential electricity prices between each program and neighboring states, relative to the year before each program started. Both programs paid more than their neighboring states, so the figure shows how that premium changed. The dashed lines show the expected change from permit costs alone, which is around 1 to 2 points for each program.

Neither program actually followed that expected path. RGGI's premium grew around 20 to 25 points between 2001 and 2008, then it fell by around 20 points after the program began. California's premium stayed near the 2012 level until 2015, then it increased rapidly by more than 40 points by 2024. That timing may point to later state-level cost pressures after the 2013 launch.

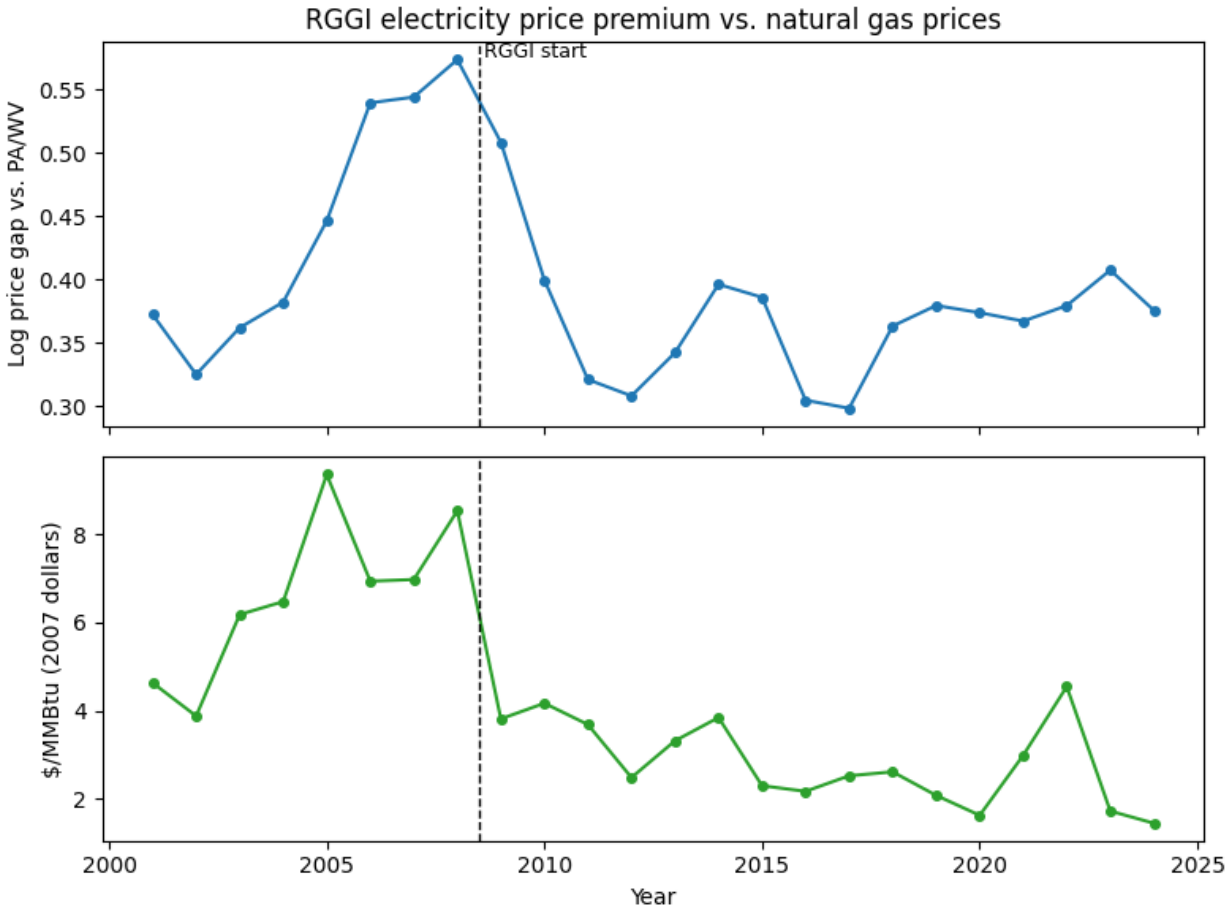
Table 2. Difference-in-differences estimates, log residential electricity price

Program	Effect (%)	Std. error	p-value	95% CI (log points)	States	n obs.
RGGI	-8.6	0.087	0.30	-0.26 to 0.08	11	264
California	16.5	0.063	0.015	0.03 to 0.28	4	96

Source: EIA residential retail electricity prices, 2001–2024, deflated with FRED series CPIAUCSL. Comparison states: Pennsylvania and West Virginia (RGGI); Oregon, Nevada, and Arizona (California). Regressions include state and year fixed effects; standard errors are clustered by state. Effect = exp(β) − 1. Author's calculations.

Table 2 shows the difference-in-differences estimates for each program. The regressions include state and year fixed effects, with standard errors clustered by state. RGGI's estimate is negative and not statistically significant. California showed an estimated increase of 16.5%, which is around eight times higher than the predicted 2% from permit costs. This effect is likely due to other state-level costs rather than the permit prices alone. Additionally, with only 4 and 11 state clusters, the standard errors are likely too small, so the p-values should not be taken as reliable.

Figure 4. RGGI vs. Henry Hub natural gas prices



Source: Price gap from EIA, average residential retail electricity price by state, 2001–2024, for the nine core RGGI states versus Pennsylvania and West Virginia. Natural gas prices are the Henry Hub spot price (FRED series MHHNGSP, originally from EIA), averaged to annual values. Both series deflated to 2007 dollars with FRED series CPIAUCSL. Author's calculations.

Figure 4 shows RGGI's premium compared to real Henry Hub natural gas prices. During the early part of the study period, RGGI states relied more on natural gas than Pennsylvania and West Virginia, which generated more energy from coal. As Table 3 shows, RGGI states generated 31% of their electricity from natural gas in 2008 compared to only 8% by Pennsylvania and 0% from West Virginia, which still heavily relied on coal. However, by 2020, Pennsylvania's natural gas share had exceeded RGGI's natural gas share, which likely weakened the link between RGGI's premium and gas prices in the later years. These two series remain relatively correlated in levels ($r = 0.64$). However, year over year, the link is weak within the

year (r = 0.25) but stronger with a one-year lag (r = 0.54). This is consistent with regulators delaying the full pass-through of costs through retail prices. Because RGGI began just as shale gas drove down natural gas prices, the data cannot separate the program's effect from the gas price shift. However, it does show that the gas price swing was far larger than what the permits could have produced.

Table 3. Share of in-state net generation by fuel source (%)

Region	Year	Coal	Natural Gas	Nuclear	Other
RGGI (9 states)	2001	23	24	28	25
	2008	22	31	29	19
	2020	1	46	30	23
Pennsylvania	2001	57	2	38	4
	2008	53	8	35	3
	2020	10	52	33	4
West Virginia	2001	98	0	0	2
	2008	98	0	0	2
	2020	89	5	0	6

Source: EIA, net generation by state, total electric power industry, 1990–2025. "Other" includes hydroelectric, petroleum, wind, solar, and biomass. Shares reflect in-state generation; RGGI states also import power. Author's calculations.

7. Limitations

There are many limitations on this preliminary analysis of the Lieberman-Warner forecasts which may play into the large discrepancies between forecasted prices and actual prices. None of the programs studied were as expansive as Lieberman-Warner would have been,

so the lower prices may partially reflect the looser caps. Additionally, Section 5 tests household effects only using state-level data from smaller programs, so it cannot confirm or rule out the 1% consumption loss Buckley and Mityakov (2009) estimate for a program as widespread as Lieberman-Warner. However, scaling California's expected effect from Table 1 based on the average permit price of $12.27 for 2015 and 2020 to the minimum price projected for 2030, the consumption loss is around 1.22%. There is also a limitation in the difference between auction prices, which were studied in this preliminary analysis, and secondary market prices for the permits, which were not studied. Finally, 2026 data only covers part of the year.

National CPI was used for every state, which may not be representative of each individual state-level economic development and health along that national time frame. California's emissions rate leaves out imported power obtained from neighboring states, not included in the data analyzed. Furthermore, the minimum detectable effect is optimistic because the yearly gaps are correlated, not independent.

The difference-in-differences comparison has its own limitations. The RGGI control group only includes Pennsylvania and West Virginia, which were the only two never-treated states that border the RGGI states directly. California's control group includes Oregon, Nevada, and Arizona. However, these states share regional power grids with the treated states, so higher permit costs may have caused the never-treated states to supply more power from their power plants. This can cause changes in the electricity prices, which cannot be completely eliminated from the way the comparison was set up. As a result, the estimated difference would likely understate the true effect. In addition, with only two or three control states, the standard errors in Table 2 are likely to be too small, so its confidence intervals are too narrow and its p-values should be read with caution. Figure 3 is therefore presented as the main visual evidence.

8. Conclusion

The 2008 Lieberman-Warner forecasts predicted much higher permit prices than any real program saw in the short term. In both 2015 and 2020, every program came in below the minimum forecasted price. Since 2020, the EU has moved into the lower end of the 2030 forecast range, though California and RGGI remain well below it. One reasonable conclusion of these results is that the forecasts may have been reasonable for the original Lieberman-Warner bill, but since the programs analyzed were looser and had experienced external events the 2008 models could not have predicted, such as cheaper fuel alternatives that produced fewer carbon emissions in the first place or the 2008 economic crisis, actual prices ended up much lower than forecast initially. The RGGI evidence shows how large the fuel shift was. RGGI's electricity premium over Pennsylvania and West Virginia followed natural gas prices, with a one-year lag, while the expected effect of permit prices stayed near 1%. At these lower prices, the programs' costs to households were too small to detect in state-level consumption or electricity prices. So what households paid for electricity was shaped more by fuel markets and state-level costs than by the price of carbon permits.

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